

JPWM06 Light reflection — Quiz 1

S3 Physics · 4 multiple-choice items

Q1 · JPWM06.2 · Foundation

The figure below shows the image of a clock formed by a plane mirror. The time is



Fig · Plane-mirror image of a clock

- A. 2:50
- B. 4:40
- C. 8:20
- D. 10:10

Q2 · JPWM06.1 · Standard

An object O is placed in front of a plane mirror and an image I is formed behind the mirror as shown in the figure below. Among rays P, Q, R and S, which of them correctly represent the reflected rays from the mirror?

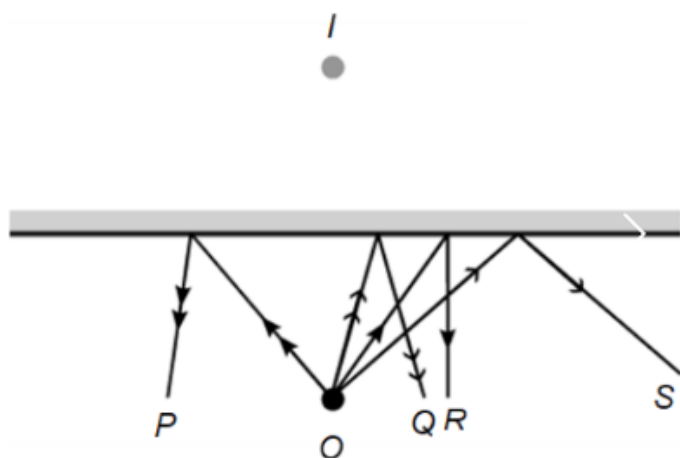


Fig · Object O and rays at a plane mirror

- A.** P and Q only
- B.** Q and S only
- C.** Q, R and S only
- D.** P, Q, R and S

Q3 · JPWM06.2 · Standard

Peter is standing 4 m in front of a plane mirror as shown. A lamp is located midway between Peter and the mirror. If Peter wants to take a clear photograph of the image of the lamp, at what distance should he focus?

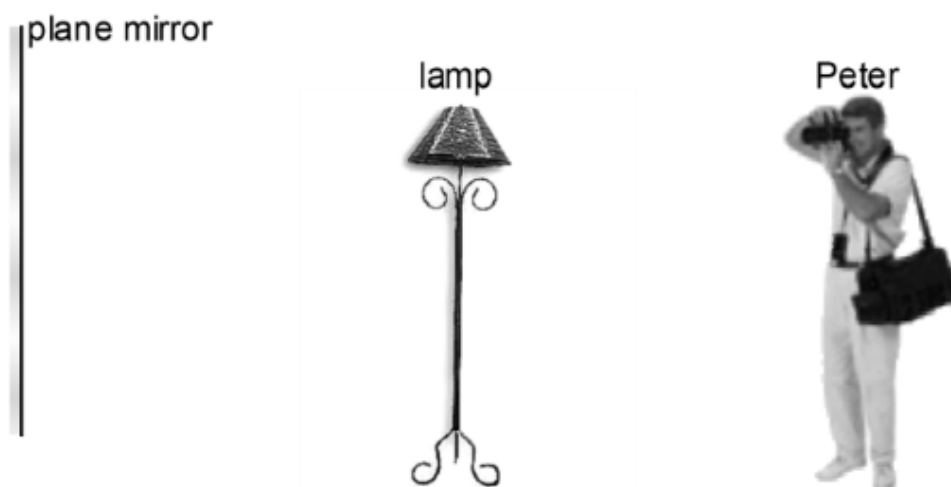


Fig · Photographing the image of the lamp

- A. 2 m
- B. 4 m
- C. 6 m
- D. 8 m

Q4 · JPWM06.1 · Standard

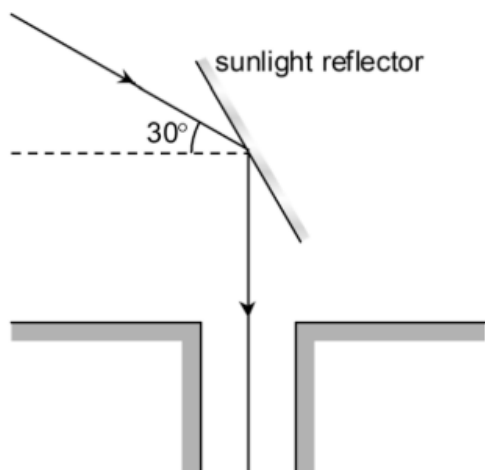


Fig · Sunlight reflector

The figure above shows a sunlight reflector installed on the roof of a building. At the instant shown, sunlight falls at an angle of 30° to the horizontal and is then reflected vertically into the building. What is the angle between the reflector and the horizontal?

- A.** 30°
- B.** 45°
- C.** 60°
- D.** 75°

JPWM06 Light reflection — Quiz 1 · Answer key

Teacher page · 4 items

Q1 · JPWM06.2

Answer: D. 10:10

Guide: A plane mirror reverses left and right. The image 12 looks like SI, which corresponds to 10:10.

Q2 · JPWM06.1

Answer: B. Q and S only

Guide: Reflected rays appear to come from the image I behind the mirror. Q and S do; P and R do not.

Q3 · JPWM06.2

Answer: C. 6 m

Guide: The lamp is 2 m from the mirror, so its image is 2 m behind the mirror. Peter is 4 m from the mirror, so he must focus 6 m away.

Q4 · JPWM06.1

Answer: C. 60°

Guide: Incident ray is 30° to the horizontal and the reflected ray is vertical. Angle of incidence equals angle of reflection, so the reflector is 60° to the horizontal.